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ORIGINAL RESEARCH ARTICLES

Aortic and Iliac Calcifications as Predictors of Aortic Dissection, Aneurysm Rupture, and Peripheral Vascular Disease: A Prospective Cohort Study from the DANCAVAS Trials

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BACKGROUND: There is limited evidence of the relationship between aortic and iliac calcification and aortic events (aortic dissection or aneurysm rupture) and major adverse limb events (MALEs; peripheral revascularization and lower limb amputation). The aim of this population-based prospective cohort study was to investigate the association of aortic and iliac calcifications with aortic events and MALEs.

METHODS: All participants from the DANCAVAS (Danish Cardiovascular Screening) trials who received a full thoracoabdominal aortic computed tomography scan with recorded measurements of aortic and iliac calcification were included. The aorta and iliac artery calcifications were measured in a 5-segment model and categorized into quartiles, apart from the ascending aorta, which was categorized into 4 exposure groups, as the majority of participants had no calcification in the ascending aorta. The participants were followed from inclusion in the DANCAVAS trial until aortic events, MALE, death, or end of follow-up on May 1, 2025. We used a competing risk hazard regression model adjusted for demographics, cardiovascular risk factors, aortic diameter, and ankle-brachial index.

RESULTS: In total, 13 065 participants were included in this study (94% men; mean age, 67±4 years). Participants were followed for a mean of 8 years, during which 42 experienced an aortic event, and 311 experienced a MALE. The analyses of the aortic event outcome showed significantly increased subdistribution hazard rates (SHRs) when having calcification scores in the fourth group or quartile for the ascending and abdominal aorta (SHR, 3.06 [95% CI, 1.15–8.16]; SHR, 7.61 [95% CI, 1.44–40.22], respectively). Calcification scores in the fourth group or quartile were strongly associated with MALEs in all segments: ascending aorta (SHR, 2.20 (95% CI, 1.56–3.10]), aortic arch (SHR, 6.49 [95% CI, 3.10–13.56]), descending aorta (SHR, 2.43; [95% CI, 1.45–4.08]), abdominal aorta (SHR, 7.89 [2.95–21.15]), and iliac arteries (SHR, 13.52 [95% CI, 4.32–42.24]).

CONCLUSION: Severe aortic and iliac calcification was associated with significantly higher risk of aortic events and MALEs, suggesting that quantifying aortic calcification may provide valuable prognostic information and potentially improve the risk stratification in patients with aortic aneurysm or peripheral artery disease.

Key Words: aortic aneurysm ■ aortic calcification ■ aortic dissection ■ cardiovascular imaging ■ major adverse limb

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Aortic dissection and aortic aneurysm rupture are life-threatening conditions with increasing incidence.[1] Major adverse limb events (MALEs), defined as peripheral revascularization or lower limb amputation, are associated with significantly reduced quality of life.[2–4]

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Clinical Perspective

What Is New?

- This study considered calcification of the entire aorta and found that severe calcification of the ascending and abdominal aorta was associated with aortic dissection and aortic aneurysm rupture.
- In this study, which is among the first of its kind, high and even modest levels of calcification in the aorta and iliac arteries were associated with major adverse limb events.

What Are the Clinical Implications?

- Aortic and iliac calcification are associated with not only major adverse limb events but also aortic diseases.
- Quantifying these calcifications could potentially be valuable input for risk stratification and guide surveillance intensity, particularly in patients with borderline aneurysm sizes or peripheral artery disease.

Nonstandard Abbreviations and Acronyms

ABI	ankle-brachial index
AU	arbitrary units
CT	computed tomography
CVD	cardiovascular disease
DANCAVAS	Danish Cardiovascular Screening Trial
MALE	major adverse limb events
MESA	Multi-ethnic study of atherosclerosis
SHR	subdistribution hazard ratio

For prevention of aortic aneurysm rupture, prophylactic surgery is recommended to patients with an aortic aneurysm mainly on the basis of the aneurysm diameter.[5] However, recent studies have investigated the relevance of ascending aortic size, indicating a need for more precise and individualized risk stratification beyond diameter alone.[6,7] As for the prevention of MALEs, screening for peripheral arterial disease with measurement of ankle-brachial index (ABI) is possible[8] but not widely implemented.

Calcifications of the coronary arteries have been extensively studied and are known to be associated with stroke and myocardial infarction.[9,10] Calcification of the aorta and iliac arteries becomes increasingly prevalent with age and shares risk factors with coronary calcification.[11,12] However, less is known about the clinical significance of aortic and iliac calcifications. Existing evidence suggests an association with stroke, myocardial infarction, and mortality,[13–17] but the links with aortic dissection, aortic aneurysm rupture, and MALEs remain insufficiently investigated.

We hypothesized that calcifications of the aorta and iliac arteries increase the risk of aortic dissection, aortic rupture, and MALEs. Accordingly, the aim of this prospective cohort study was to investigate the association of aortic and iliac calcification with aortic and peripheral diseases using 8-year follow-up data from the DANCAVAS (Danish cardiovascular screening) Trials.

METHODS

Setting

This study was a population-based prospective cohort study. Data for this study were obtained from the DANCAVAS trials I and II, which were population-based, randomized, multicenter trials conducted at 5 Danish hospitals (Odense University Hospital, Svendborg Hospital, Vejle Hospital, Silkeborg Hospital, and Nykøbing Falster Hospital) from September 2014 to February 2019.[18,19]

The participants were identified from the Danish Civil Registration System and resided in the catchment area of the participating hospitals. They were randomized at a 1:2 ratio (DANCAVAS I) and a 1:4 ratio (DANCAVAS II) to either receive an invitation for advanced cardiovascular screening or to continue with usual medical care. Of the invited participants, about two-thirds accepted the invitation, resulting in approximately 15 000 individuals aged 60 to 75 years attending the screening program. Both women and men were enrolled in the pilot phase of the study; however, from May 2015, only men were included. For more details on the DANCAVAS trials, the study protocol can be accessed with trial registration numbers NCT03946410 and ISRCTN12157806.

The observation period for this study started on the date of inclusion in the DANCAVAS trials. The participants were then followed until first event, emigration, death, or end of follow-up on May 1, 2025.

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Participants

This study population derives from the group of screened participants in the DANCAVAS trials. All screened participants with a computed tomography (CT) scan and measurement of at least 1 aortic or iliac calcification score were eligible for inclusion. There were no further exclusion criteria for this study.

Approval

Participants provided written and verbal informed consent upon enrollment in the DANCAVAS trials. The Danish Data Protection Agency and the Regional Scientific Ethical Committee of Southern Denmark (S-20140028, 2014) approved the studies.

Data Sharing Statement

Due to the inclusion of individual-level data from national registries, the authors are unable to share or grant direct access to the dataset. However, licensed institutions interested in accessing the data may submit an application to the Danish Health Data Authorities in accordance with regulatory requirements. C.S.P. has full access to all data and takes responsibility for their integrity and analysis.

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Data were masked if a single cell had fewer than 5 participants (n≤5) according to regulation from the Danish Health Data Authority.

Data Collection

All participants completed a baseline questionnaire about prior cardiovascular disease (CVD), family history of CVD, smoking status, medications (including antiplatelet, anticoagulant, lipid-lowering, antihypertensive. and antidiabetic agents), heart disease symptoms, and symptoms of peripheral artery disease. Additionally, the following measurements were taken: weight, height, and blood pressure (on upper and lower extremities), and blood samples were drawn. Body mass index, 10-year atherosclerotic CVD risk score,[20,21] and ABI were calculated. See Table S1 for detailed information on variable definitions.

Follow-up data on events were obtained from the following Danish health registries: the Danish Civil Registration System[22] (providing information on emigration and time of death) and the Danish National Patient Registry[23,24] (containing data on all hospital admissions, including time and date, diagnosis codes using the *International Classification of Diseases* system, and procedures coded with the Nordic Classification of Surgical Procedures codes). Complete follow-up data on all participants were obtained.

Cardiac Noncontrast CT Scan

All participants received an electrocardiogram-gated noncontrast CT scan upon inclusion in the DANCAVAS trials; see Table S2 for more information on the scanners used. Scanner settings depended on scanner and participant characteristics according to guidelines. The reconstructed slice thickness was typically 3 mm. All participants were scanned from the mandibular bone to the proximal third of the femur.

Measurement of Calcifications and Diameters

The aortic and iliac calcification scores were measured using a software based on the Agatston method.[25] Experienced technicians blinded to participant data continuously analyzed the aortic and iliac calcification scores. Methods S1 provides additional details on the methodology for generation of calcification scores.

For measurement of calcification scores, the aorta and iliac arteries were divided into the following segments: ascending aorta, aortic arch, descending aorta, suprarenal aorta, infrarenal aorta, common iliac arteries (arteria iliac communis dextra et sinistra), and external iliac arteries (arteria iliaca externa dextra et sinistra); see Figure 1 for a visual illustration of the segment division.

The segments were measured as follows: (1) the ascending aorta from the separation of the aortic arch to the separation of the pulmonary trunk, (2) the aortic arch from just below the branching of the subclavian and carotid arteries to the level where the aortic arch separates into the ascending and descending aorta, (3) the descending aorta from the aortic arch ending to the insertion of the diaphragm onto the aorta, (4) the suprarenal aorta from the diaphragm to where both the left and right renal artery had branched, (5) the infrarenal aorta from the branching of the renal arteries to the aortic bifurcation, (6) the common iliac arteries from the aortic bifurcation to the bifurcation into internal and external iliac arteries, and (7) the external iliac arteries from the ending of the common iliac arteries to the superior border of the femoral head. For the statistical analysis, the participants were divided into 4 a priori-defined quartiles on the basis of calcification score quartiles for the aortic arch, descending aorta, abdominal aorta, and iliac arteries (the abdominal aorta comprised the suprarenal and infrarenal aorta, and the iliac arteries comprised the bilateral common and external iliac arteries). Due to right skewness in calcification scores of the ascending aorta, 4 manually defined groups were used instead of quartiles to allow for more meaningful categorization. The groups were as follows: Agatston score 0 arbitrary units (AU), 1 to 9 AU, 10 to 99 AU, and ≥100 AU. Methods S2 provides additional details on group definition.

Additionally, aortic diameters were obtained, measured from the exterior to the exterior surface of the aortic wall, as stated in guidelines[26]; see Table S1 for a description of how the diameter measurements were performed. If any dilatation was found, then the diameter was instead measured at the site of the dilatation. Measurements of iliac diameter were only obtained when an iliac aneurysm was suspected (diameter >20 mm).

Outcomes

The outcomes were the composite outcomes of aortic events and MALEs. The outcome of aortic events comprised aortic dissection and aortic aneurysm rupture at any aortic location. The MALE outcome included peripheral revascularization and amputation of a lower limb; see Table S3 for the *International Classification of Diseases*, *Tenth Revision* codes for both outcomes.

Statistical Methods

Calcifications in the ascending aorta, aortic arch, descending aorta, abdominal aorta, and iliac arteries were statistically analyzed separately, and each segment was analyzed for both outcomes (aortic events and MALEs). Participants were excluded from the analysis of a particular segment if the segment had missing calcification scores.

A competing risk hazard regression model was used to examine the association between aortic and iliac calcification and aortic events and MALEs. We conducted 2 separate analyses, 1 for aortic events (composite end point of aortic dissection and aneurysm rupture) and one for MALEs (composite end point of peripheral revascularization and lower limb amputation). In the analysis of aortic events, elective aortic aneurysm surgery and death were treated as competing risks; for MALE, only death was considered a competing risk. The risk period began at the date of inclusion and ended at the earliest of the events (aortic event or MALE), death, emigration, or end of follow-up on May 1, 2025. The proportional hazard assumption was assessed with Schoenfeld residuals and visual inspection of log-log plots of outcome versus analysis of time. Incidence rates are presented in the relevant tables.

The data were analyzed using 4 regression models, a crude model, 2 minimally adjusted models (first, age and sex; second, age, sex and diameter), and a fully adjusted model. The fully adjusted model included age, sex, body mass index, smoking status, hypertension, diabetes, atrial fibrillation, LDL (low-density lipoprotein) cholesterol, systolic blood pressure, estimated glomerular filtration rate, and self-reported myocardial infarction, stroke, chronic obstructive pulmonary disease, use of

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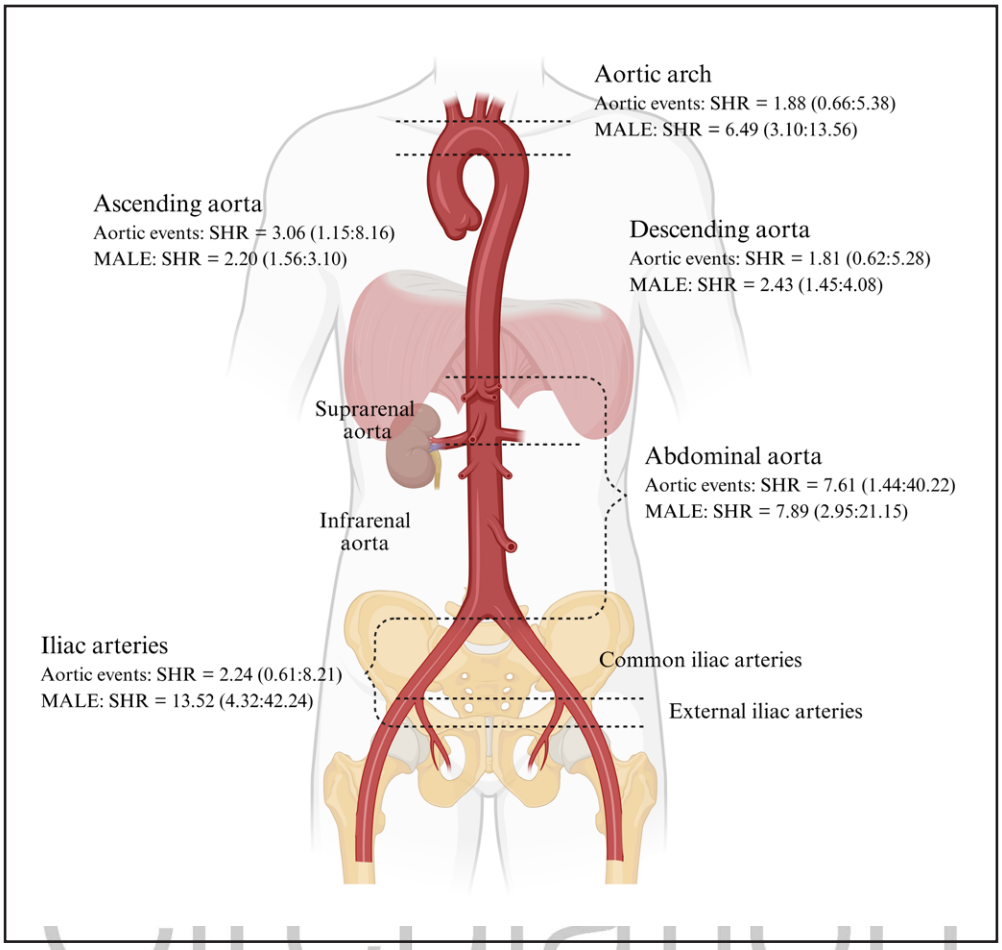

Figure 1. Illustration of the division of the aorta and iliac arteries to obtain calcification scores and the associated subdistribution hazard ratios (SHRs; ascending aorta fourth group and fourth quartile for the remaining).
The dotted lines indicate the segment borders. The hazard ratios presented are for the fourth group and quartile. Abdominal aorta, suprarenal and infrarenal aortae; iliac arteries, the common iliac and external iliac arteries; aortic events, aortic dissection and aortic aneurysm rupture; major adverse limb event (MALE), peripheral revascularization and lower limb amputation.

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statin therapy, hereditary predispositions, and the diameter of the aortic segment in question. In the analysis of the iliac segment, the adjusted model also included iliac diameter, and ABI was included in all models on MALEs. Additionally, 2 post hoc analyses were performed; 1 stratified for sex and 1 pooling the first, second, and third groups/quartiles and comparing these combined strata with the fourth group/quartile. Information and details on variable definitions and data sources and how they were treated can be found in Table S1.

The first group of the ascending aorta and the first quartile of the remaining segments were the reference categories. The correlation between aortic diameter and calcification score was assessed by inspecting scatterplots, and potential relationships were evaluated using Spearman's rho. Furthermore, the association between the number of segments with high calcium scores and outcomes were examined.

We performed a sensitivity analysis, investigating SHRs when elective aortic aneurysm surgery was included in the aortic events outcome. Thus, the outcome in this sensitivity analysis comprised aortic dissection, aneurysm rupture, and elective aortic aneurysm surgery. This regression was adjusted for the same variables as the adjusted regression model in the main analysis.

An additional sensitivity analysis was conducted on participants with no prior CVD, which included self-reported myocardial infarction, bypass surgery, percutaneous coronary intervention, heart valve surgery, and stroke. The sensitivity analysis was a competing risk hazard regression model adjusted as mentioned above, with the exception of prior CVD.

A sensitivity analysis was done on continuous calcification scores for all 5 segments. To account for the large number of 0 values, this analysis was performed with a log2(+0.5) transformation of values.

The statistical analyses were performed in STATA/IC 19.0 (StataCorp, College Station, TX). $P<0.05$ was considered statistically significant. All continuous variables were displayed as mean (SD), categorical variables were displayed as numbers (percentages), and SHRs were presented together with 95% CIs.

RESULTS

Baseline Data

From September 25, 2014 to February 27, 2019, 15 003 participants were included in the DANCAVAS trials at the

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5 inclusion sites. However, only 13 065 participants met the inclusion criteria of having at least 1 measured calcification score in the aorta or iliac arteries (Figure 2). No noteworthy differences were seen between the included and excluded participants (Table 1). The population consisted of 743 (5.7%) women and 12 322 (94.3%) men aged 60 to 75 years with a mean age of 67.3 years (3.8 years); further baseline characteristics are shown in Table 1.

The population had a very high prevalence of aortic and iliac calcification, with 99.6% of the participants having a score above 0 AU in at least 1 segment and 39.5% with a score above 0 AU in all 5 segments. Margins for calcification scores in each segment are presented in Table 2. Calcification scores were highest in the abdominal aorta, with a median score of 1993 AU.

Follow-Up and Events

The mean follow-up time was 7.7 years (1.7). Of all participants, 42 participants experienced an aortic event, and 311 participants experienced a MALE. Table S4 presents the number of events in each group or quartile for both outcomes.

Main Results

Aortic Events (Combined End Point of Aortic Dissection and Aneurysm Rupture)

SHRs from the adjusted regression model for both outcomes can be found in Figure 3. The analyses showed that calcification scores in the fourth group of the ascending aorta and the fourth quartile of the abdominal aorta were associated with increased SHR for aortic events (SHR, 3.06 [95% CI, 1.15–8.16]; SHR, 7.61 [95% CI, 1.44–40.22], respectively). Furthermore, in the ascending and abdominal aorta for each 1-mm increase in diameter, the hazard for the event increased by 18% (SHR, 1.18 [95% CI, 1.08–1.28]) and 6% (SHR, 1.06 [95% CI, 1.03–1.08]), respectively; see Table S5A through S5E for further details. Calcification scores in the aortic arch, descending aorta, and iliac arteries had no association with aortic events. See Table S6A and S6B for hazard ratios from the crude regression model and the 2 minimally adjusted models.

When inspecting scatterplots for any potential correlation between aortic diameter and aortic calcifications, we found no correlation; see Table S7 for calculated Spearman's rho. The association between number of segments with high calcium scores and outcomes are reported in Table S8A and S8B. It revealed that increasing numbers

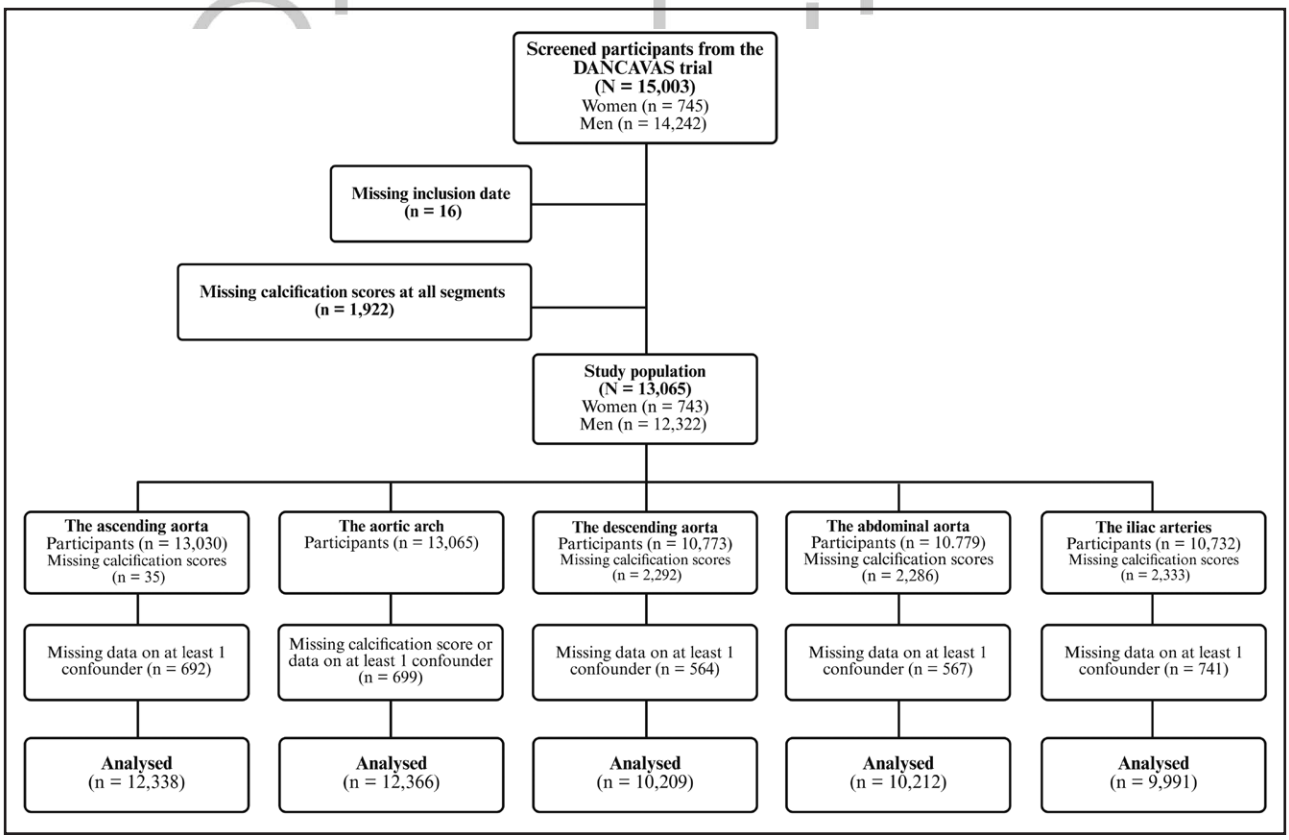

Figure 2. Flowchart of the inclusion and exclusion of participants in this prospective cohort study investigating the association of aortic and iliac calcification with aortic events and major adverse limb events.
Due to the masking request from the Danish Health Data Authority, the number of missing calcification scores is not presented for the aortic arch segment. DANCAVAS indicates Danish cardiovascular screening trial.

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Table 1. Baseline Characteristics of the Study Population Derived From the Danish Cardiovascular Screening Trials and the Participants Who Were Excluded due to Missing Calcification Scores

	Study population (n=13 065)	Missing calcification score (n=1922)
Age, y	67.3 (3.8)	67.6 (3.5)
Sex, men, %	94.3	99.9
Height, cm	176.7 (7.3)	176.1 (6.3)
Weight, kg	87.7 (15.3)	86.7 (14.9)
Body mass index, kg/m^2	28.1 (4.4)	27.9 (4.4)
Systolic blood pressure, mm Hg	150.1 (18.7)	143.8 (17.9)
Diastolic blood pressure, mm Hg	82.9 (9.9)	79.8 (9.3)
LDL cholesterol, mg/dL	2.9 (1.0)	2.8 (1.0)
Estimated glomerular filtration rate, mL/min/1.73 m^2	78.0 (13.7)	79.8 (13.3)
10-year ASCVD risk categories, n (%)		
Low-risk (<5%)	7 (0.05)	1 (0.052)
Borderline risk (5% to <7.5%)	108 (0.83)	21 (1.09)
Intermediate risk (7.5% to <20%)	4077(31.21)	669 (34.81)
High risk (≥20%)	8646 (66.17)	1212 (63.06)
Smoking status, n (%)		
Never	4399 (33.8)	674 (35.2)
Former	6508 (50.0)	957 (50.0)
Current	2105 (16.2)	284 (14.8)
Comorbidities, n (%)		
Diabetes	1656 (12.8)	226 (11.8)
Hypertension	6966 (53.5)	839 (43.7)
Chronic obstructive pulmonary disorder	787 (6.0)	117 (6.1)
Aortic aneurysm	207 (1.6)	34 (1.8)
Stroke	846 (6.5)	112 (5.8)
Myocardial infarction	764 (5.8)	104 (5.4)
Atrial fibrillation	1848 (14.2)	236 (12.3)
Medical treatment, n (%)		
Platelet inhibitors	2758 (21.2)	417 (21.5)
Anticoagulant therapy	875 (6.7)	143 (7.4)
Statin therapy	4317 (33.0)	644 (33.2)
Antihypertensive therapy	6156 (47.1)	891 (46.0)

All variables are displayed as mean (SD) unless otherwise stated. Comorbidities and medical treatments were self-reported and are presented as number of participants (the percentage of participants) with the given comorbidity or medical treatment. ASCVD indicates atherosclerotic cardiovascular disease and LDL, low-density lipoprotein.

of segments with severe calcifications were associated with increasing aortic events and MALEs, respectively.

MALEs (Combined End Point of Peripheral Revascularization and Lower Limb Amputation)

In the aortic arch and abdominal aorta, calcification scores in the second quartile or above were associated with significantly increased SHRs for MALEs (fourth quartile, 6.49 [95% CI, 3.10–13.56]; fourth quartile, 7.89 [95% CI, 2.95–21.15]), respectively). Furthermore, having calcification scores in the fourth group of the ascending aorta (2.20 [95% CI, 1.56–3.10]), the fourth quartile of the descending aorta (2.43 [95% CI, 1.45–4.08]), or the third or fourth quartile of the iliac arteries (fourth quartile, 13.52 [95% CI, 4.32–42.24]) were likewise associated with increased hazard rates in the adjusted model. SHRs for the covariates can be found in Table S5A through S5E. See Figure 1 for an illustrative summary of the aortic and iliac segments and SHR in the fourth group or quartiles.

Table S9 shows the results from the post hoc stratification of the main analysis. However, it was only possible to stratify for men due to a very limited number of events among women. There were no noteworthy differences compared with the main results.

Table S10 reports the results from the second post hoc analysis, pooling the first, second, and third strata. In the pooled analysis for aortic events, we observed a significantly increased risk of events when having calcification scores in the fourth group/quartile of the ascending and abdominal aorta, similar to the findings of the main analysis. Likewise, in the pooled analysis of MALEs, significant hazard rates were observed in the same groups and quartiles as in the main analysis.

Sensitivity Analyses

Figure 4 presents the results from the sensitivity analysis where the aortic outcome comprised aortic dissection, aneurysm rupture, and elective aortic aneurysm surgery (resulting in 144 aortic events). This analysis revealed significant associations between calcification scores in the fourth group of the ascending aorta (1.79 [95% CI, 1.08–2.96]) and in the fourth quartile of the abdominal aorta (2.91, [95% CI, 1.18–7.17]), which corresponds with the findings in the main analysis. Additionally, the analysis showed a statistically significant association between calcification scores in the fourth quartile of the descending aorta and iliac arteries with an SHR of 2.26 (95% CI, 1.20–4.28) and 3.14 (95% CI, 1.45–6.81), respectively.

Hazard ratios from the sensitivity analysis excluding those with prior CVD (17% of the study population) are presented in Table 3. This analysis found no association between calcifications of the aorta and iliac arteries and aortic events; however, this was not the case for MALEs. Calcification scores in the third and fourth group of the ascending aorta and the third and fourth quartile of the aortic arch, abdominal aorta, and iliac arteries were significantly associated with increasing hazard rates for MALEs. For the descending aorta, only calcification scores in the fourth quartile had a significant association with MALEs.

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Table 2. Margins for Calcification Scores in the Aorta and Iliac Arteries by Group and Quartile, Stratified by Sex Together With Event Counts for Each Event Type, the Incidence Rates, and the Diameter of the Aortic Segment and the Number of Missing Calcification Scores of Each Segment

Segment	First group	Second group	Third group	Fourth group	Aortic events		MALE		d	Missing
					(n)	IR (95% CI)	(n)	IR (95% CI)	(mm)	(n)
Ascending, n	8681	1363	1363	1623						
Women (n=743)	0	1–9	10–99	100–18 417	≤5	0.43 (0.14–1.34)	8	1.15 (0.58–2.30)	34.1	≤5
Men (n=12 287)	0	1–9	10–99	100–14 567	34	1.34 (1.13–1.60)	264	2.83 (2.51–3.20)	37.4	35
	Q1	Q2	Q3	Q4						
Arcus, n	3301	3230	3266	3265	…	…	…	…	…	…
Women (n=743)	0–23	24–246	247–893	894–11–096	≤5	0.43 (0.14–1.34)	8	1.15 (0.58–2.30)	28.2	≤5
Men (n=12 319)	0–20	21–224	225–804	805–233 223	34	1.34 (1.12–1.59)	265	2.84 (2.52–3.20)	30.7	≤5
Descending, n	2858	2543	n=2681	n=2691						
Women (n=743)	0–7	8–105	106–657	658–35 114	≤5	0.43 (0.14–1.34)	8	1.15 (0.58–2.30)	25.3	≤5
Men (n=10 030)	0–7	8–55	56–478	479–42 332	30	1.36 (1.12–1.65)	219	2.88 (2.53–3.29)	28.5	2292
Abdominal, n	2695	2698	2692	2694						
Women (n=742)	0–265	266–1233	1234–3897	3898–20 789	≤5	0.43 (0.14–1.34)	8	1.15 (0.58–2.31)	16.7	≤5
Men (n=10 037)	0–515	516–2053	2054–5684	5685–130 259	30	1.36 (1.12–1.65)	217	2.85 (2.50–3.26)	20.5	2285
Iliac, n	2694	2674	2681	2683						
Women (n=737)	0–14	15–372	373–1698	1699–19 622	≤5	0.43 (0.14–1.35)	8	1.15 (0.58–2.32)	…	6
Men (n=9995)	0–220	221–1146	1147–3392	3394–44 748	30	1.37 (1.13–1.66)	217	2.87 (2.51–3.27)	…	2327

Margins for calcification scores are stratified by sex in the aorta and iliac arteries by group or quartile. Calcification scores are presented in arbitrary units. Incidence rates are presented as risk/1000 person-years. Diameters (d) are mean (SD). Data are masked when a single cell has fewer than 5 counts, as required by regulations from the Danish Health Data Authority. IR indicates incidence rate and Q, quartile. Data is masked when a single cell has fewer than five counts (n ≤ 5) as required by regulations from the Danish Health Data Authority

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Results from the sensitivity analysis of continuous calcification are reported in Table S11. For aortic events, we observed significant coefficients for the ascending and abdominal aorta. In these segments, a doubling in calcification score increased the risk of event by 12% (95% CI, 1.01–1.25) and 29% (95% CI, 1.05–1.59), respectively. The analyses of MALEs revealed that a doubling of calcification score in all 5 segments significantly increased the risk of events.

DISCUSSION

This large population-based prospective cohort study revealed a clear association between high levels of aortic and iliac calcification and the risk of major vascular events. In the competing risk analyses for aortic events, individuals with calcification scores in the fourth group and fourth quartile of the ascending and abdominal aorta showed significant hazard ratios. This relationship was even more pronounced for MALEs, where substantially elevated hazard rates were observed for calcification scores in the fourth group or fourth quartile across all aortic and iliac segments.

Based on existing literature, an association between calcifications of the aorta and aortic events was anticipated. Over the last decades, several population-based cohort studies have investigated aortic calcification as a risk factor for coronary heart disease and death.[14,17,27] However, to our knowledge, no other studies have investigated the association of calcifications of the entire aorta with aortic events. The Rotterdam study[17] has linked severe aortic arch calcification with increased risk of all-cause mortality. In MESA (Multi-Ethnic Study of Atherosclerosis),[14] they found significant associations between abdominal aortic calcifications and CVD (including coronary heart disease, stroke, and cardiovascular death). Both of these population-based studies presented results similar to ours, supporting the theory that aortic calcifications are associated with the occurrence of these adverse cardiovascular events.

The pathophysiology behind calcifications of the aorta and the risk of cardiovascular events is being investigated. One study[28] has identified increased vascular calcification in aortic aneurysms, and another study[29] has observed increased amounts of calcification in aortic tissue more disposed to failure. The findings in these studies suggest that arterial calcifications weakens the vessel wall, making it more prone to rupture. However, the mechanisms are far from fully understood.

In contrast to these findings, Chau et al.[30] found a negative association between ascending aortic aneurysms and aortic dissection and atherosclerosis, concluding that ascending aortic aneurysms protect against calcifications. The contrasting conclusions could be due to the fact that Chau et al. only considered type A

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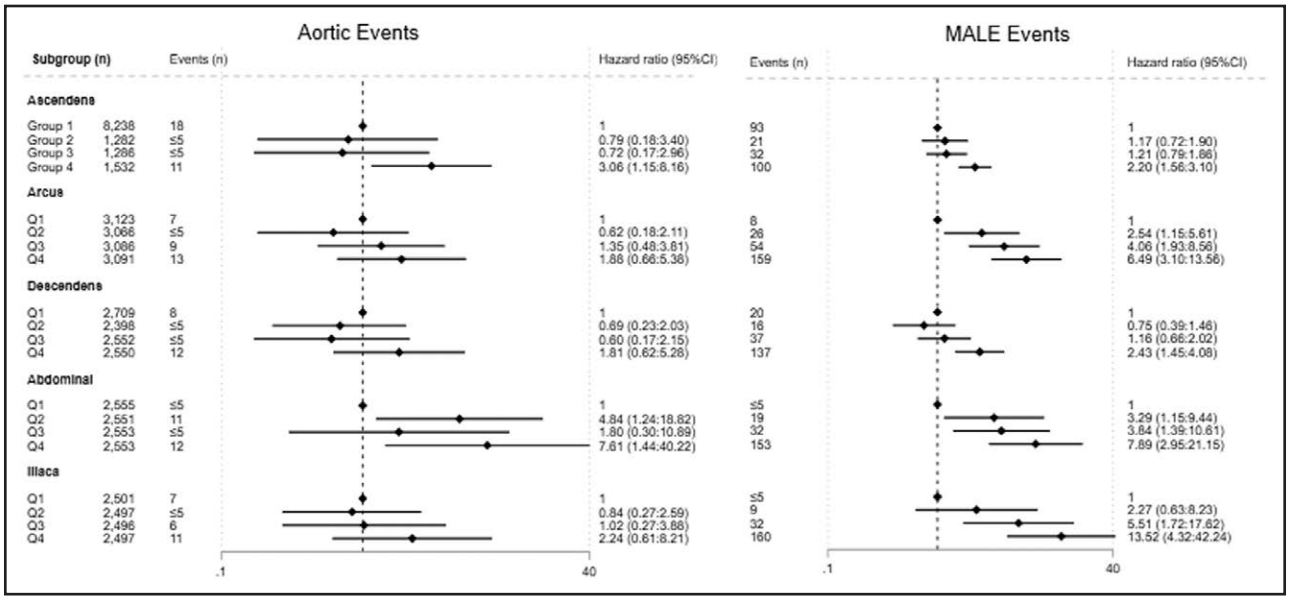

Figure 3. Forest plot displaying the number of participants, number of events, and hazard ratios for the adjusted regression models investigating the association of aortic and iliac calcifications with aortic events and major adverse limb events in a population of 13 065 participants.

Aortic events, aortic dissection and aortic aneurysm rupture; major adverse limb event (MALE), peripheral revascularization and lower limb amputation. The regression models were adjusted for sex, age, body mass index, low-density lipoprotein cholesterol, systolic blood pressure, estimated glomerular filtration rate, smoking status, comorbidities, statin treatment, hereditary predisposition, and the diameter of the aortic segment in question. In the iliac segment and for major adverse limb events, the model also included the ankle-brachial index. Data are masked when a single cell has fewer than 5 counts, as required by regulations from the Danish Health Data Authority. Q indicates quartile.

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dissection, whereas we included type A and B dissection and aneurysm rupture in our composite aortic outcome. Another explanation for the differences could be that the ascending aorta is generally less calcified compared with the other aortic segments. Furthermore, it is worth considering the small study population and the study design used by Chau et al.. These conflicting conclusions highlight the need to investigate this field even further to better understand the role of aortic calcifications in aortic dissection.

Adjustment for traditional cardiovascular risk factors in the regression models further strengthened the observed associations between aortic events and calcifications in both the ascending and abdominal aorta. In addition, we adjusted for the diameter based on a suspicion that a greater diameter potentially contained a larger amount of calcifications due to the greater vessel surface. Of note, we found no correlation upon investigating the correlation between diameter and calcification score in the aortic segments. This finding was similar to those in existing literature,[31,32] making aortic calcifications independent of diameter.

In our first sensitivity analysis, we extended the composite aortic event to include elective aortic aneurysm surgery, as it is performed to prevent rupture of aortic aneurysms. This analysis disclosed significant relationships between not only aortic events and the ascending and abdominal aorta but also the descending aorta and iliac arteries. The participants in the DANCAVAS trials were screened at baseline, and potential aortic aneurysms were discovered and treated, thus preventing aneurysm rupture. Therefore, it was important to include elective aneurysm surgery in the regression models to get the full picture of the risk and not underestimate the significance of aortic calcification.

Furthermore, we found a strong association between aortic and iliac calcifications and MALEs, even after adjustment for ABI. The opportunity to adjust for ABI offers unique insight into aortic calcifications as a risk factor for MALEs. The connection between aortic calcification and MALEs is sparsely explored, with few studies done in this area. Our expectation was to find a relationship between calcifications of the iliac arteries and MALEs, as peripheral arterial calcification can result in worsening of limb ischemia and often precedes revascularization and amputation.[33,34] What was unexpected was the great association between calcifications of the ascending aorta, aortic arch and descending aorta, and MALEs. This association continued even after the exclusion of participants with prior CVD, and most surprisingly, the ascending aorta remained significantly associated with MALE in this analysis, suggesting that calcification of the aorta is an expression of more systemic calcification, implying that, when calcifications have formed in the aorta, a substantial amount of calcification is already present in the peripheral arteries.

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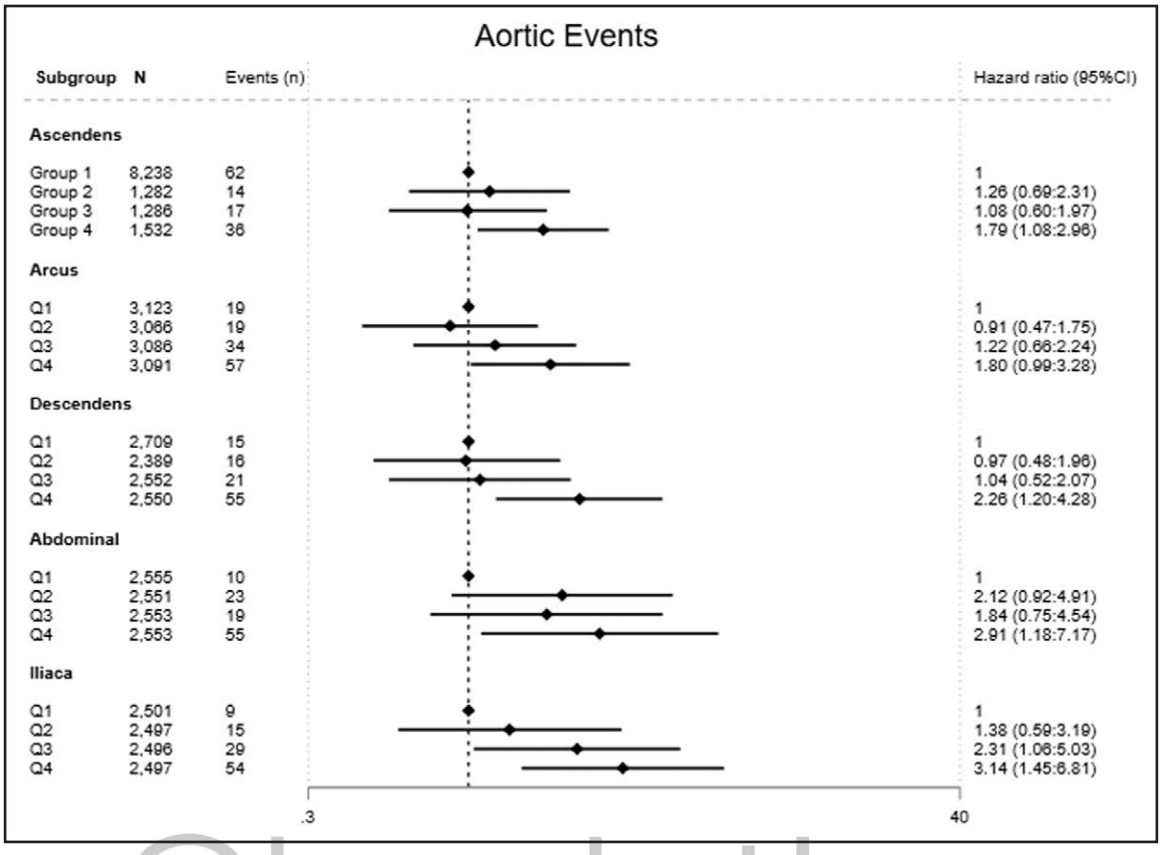

Figure 4. Forest plot displaying the number of participants, number of events, and hazard ratios for the adjusted regression models investigating the association of aortic and iliac calcifications with aortic events (including elective aortic aneurysm surgery).

Aortic events, aortic dissection, aortic aneurysm rupture, and elective aortic aneurysm surgery. The regression models were adjusted for sex, age, body mass index, low-density lipoprotein cholesterol, systolic blood pressure, estimated glomerular filtration rate, smoking status, comorbidities, statin treatment, hereditary predisposition, and the diameter of the aortic segment in question. Q indicates quartile.

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Strengths and Limitations

The calcification scores were obtained from CT scans using the Agatston method, which is the gold standard. With inspiration from previous studies,[17,35,36] we aimed to divide our population into quartiles based on calcifications scores but were unable to do so for the ascending aorta. The calcification scores of the common femoral arteries were not measured. This is a limitation to the study, as the inclusion of common femoral artery calcification would be highly relevant when investigating the MALE outcome. Other studies[17,36] measured the calcification score of the aortic arch in an age group comparable with ours and found median calcification scores similar to those found in this study. This suggests that aortic calcification measured on CT scans using the Agatston method can be suitable for larger-scale use. Measuring aortic calcification using the Agatston method gives an easily reproducible output with low interobserver variations similar to the coronary artery calcium score, which is highly reproducible.[37]

Of the abovementioned population-based studies,[14,17,27] ours includes the largest study population when investigating aortic and iliac calcifications. The size of the population improves the estimates and gives greater insight into the significance of aortic calcification. However, as we mostly included elderly men, our findings cannot be generalized to women and younger persons. Although, calcifications are most prevalent among men,[38,39] future studies should include women, as the disease burden among women is non-negligible.[40]

The Danish health registries have high validity and are known to provide detailed information on an individual level.[41,42] By using these registries, we were able to obtain full information on all diagnoses and hospitalizations of our participants. This gives us precise follow-up data and greatly reduces the risk of information bias regarding our event. Unfortunately, regulations from the Danish Health Data Authority prevent us from reporting exact numbers, which is why figure and table cells with fewer than 5 observations are masked.

The covariates adjusted for were partly self-reported, which introduced the risk of recall and information bias. Another limitations is the vast majority of White descent in the DANCAVAS trial (96% of the participants), thus the results may not be applicable to other ethnicities.[19]

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Table 3. Results From the Sensitivity Analysis Excluding Participants With Prior Cardiovascular Disease When Investigating the Association of Calcification Scores With Aortic Events and Major Adverse Limb Events

Aortic segment	Aortic events		MALE	
Ascending	SHR (95% CI)	Events (n)	SHR (95% CI)	Events (n)
First group (n=7147)	1	17	1	57
Second group (n=1090)	0.49 (0.06–4.01)	≤5	1.37 (0.74–2.53)	14
Third group (n=1010)	0.73 (0.18–2.95)	≤5	1.90 (1.12–3.20)	24
Fourth group (n=1091)	3.09 (0.98–9.73)	7	3.17 (2.03–4.93)	68
Arcus				
Q1 (n=2845)	1	7	1	8
Q2 (n=2652)	0.54 (0.13–2.25)	≤5	1.70 (0.69–4.20)	16
Q3 (n=2587)	0.85 (0.23–3.16)	6	3.49 (1.55–7.83)	39
Q4 (n=2200)	2.36 (0.76–7.39)	11	5.92 (2.68–13.05)	100
Descending				
Q1 (n=2433)	1	7	1	12
Q2 (n=2144)	0.77 (0.26–2.33)	6	0.73 (0.31–1.75)	11
Q3 (n=2073)	0.32 (0.06–1.63)	≤5	1.50 (0.75–3.00)	30
Q4 (n=1808)	1.49 (0.45–4.96)	9	2.94 (1.53–5.64)	76
Abdominal				
Q1 (n=2340)	1	≤5	1	≤5
Q2 (n=2261)	4.15 (1.00–17.26)	10	2.83 (0.81–9.96)	13
Q3 (n=2102)	1.30 (0.18–9.49)	≤5	3.82 (1.19–12.26)	21
Q4 (n=1761)	6.21 (0.91–42.19)	9	8.57 (2.76–26.57)	90
Iliac				
Q1 (n=2305)	1	6	1	≤5
Q2 (n=2184)	0.78 (0.23–2.68)	≤5	2.09 (0.41–10.63)	6
Q3 (n=2052)	1.15 (0.29–4.51)	6	4.81 (1.12–20.56)	17
Q4 (n=1743)	2.21 (0.52–9.38)	8	17.37 (4.25–70.97)	102

Aortic events are aortic dissection and aortic aneurysm rupture. Major adverse limb events (MALEs) are peripheral revascularization and lower limb amputation. Prior cardiovascular disease was defined as prior stroke, myocardial infarction, bypass surgery, percutaneous coronary intervention, or heart valve surgery. The regression models were adjusted for sex, age, body mass index, low-density lipoprotein cholesterol, systolic blood pressure, estimated glomerular filtration rate, smoking status, comorbidities, statin treatment, hereditary predisposition, and the diameter of the aortic segment in question. In the iliac segment and for MALEs, the model also included the ankle-brachial index. First group, 0 arbitrary units; second group, 1 to 9 AU; third group, 10 to 99 AU; and fourth group, ≥100 AU. Data are masked when a single cell has fewer than 5 counts, as required by regulations from the Danish Health Data Authority. Q indicates quartile and SHR, subdistribution hazard ratio.

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The validation of DANCAVAS data is described in the DANCAVAS study protocol.[43]

Perspectives

Our findings suggest that quantifying aortic calcification may offer important prognostic information about the risk of aortic dissection, aortic aneurysm rupture, and MALEs. This may potentially improve risk stratification and guide surveillance intensity, particularly in patients with borderline aneurysm sizes or peripheral artery disease.

Furthermore, with the growing adoption of artificial intelligence in radiology, automated detection and quantification of aortic and iliac calcification may enable opportunistic risk assessment from routine CT scans.

However, further research is needed to determine whether systematic reporting of aortic calcification should be integrated into standard radiology workflows to support earlier identification and management of high-risk patients.

Conclusion

In this population-based cohort study including primarily men aged 60 to 75 years, we found statistically significant associations between severe levels of aortic and iliac calcification and the risk of aortic events and MALEs.

ARTICLE INFORMATION

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Disclosures

None.

Supplemental Material

Methods S1
Methods S2
Tables S1–S10

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